updated: 4/26

Curriculum Vitae: Chad McKell

ABOUT

Title	Research Scientist | Numerical Acoustics & Computational Audio
Company	Sigma Connectivity, on assignment at Meta
Address	10301 Willows Road NE, Redmond, WA 98052
Email	`cmckell@meta.com`
Website	`chadmckell.com`
Languages	English (native), Spanish (B2), French (A2)
Code	Python, C++/CUDA, MATLAB, LaTeX, Java, HTML
Research	acoustic wave simulation and synthetic data engineering for audio AI

EDUCATION

(exp. 2026)	**University of California San Diego**, Ph.D. in Computer Music Research area: Computational Acoustics and Applied Mathematics Dissertation: *Conformal Symmetry for Exterior Wave Problems* Co-advisors: Albert Chern (CSE), Miller Puckette (Music; IRCAM)
2017	**University of Edinburgh**, M.S. in Acoustics and Music Technology Research area: Computational Acoustics. Advisor: Stefan Bilbao.
2015	**Wake Forest University**, M.S. in Physics
2009	**Brigham Young University**, B.S. in Biophysics

EMPLOYMENT

9/24–	**Sigma Connectivity, on assignment at Meta**, Research Scientist
3/24–6/24	**University of California San Diego**, Teaching Assistant in Computer Science
6/23–2/24	**Meta, Reality Labs Research**, Research Scientist Intern
8/21–3/22	**Meta, Reality Labs Research**, Research Intern
7/18–7/19	**Applied Research in Acoustics**, R&D Scientist
4/17–9/17	**Lofelt**, Research Consultant – Acquired by Meta in 2022
10/14–8/16	**J.P. Morgan/Neovest (via ConsultNet)**, SDET
9/09–9/11	**Wake Forest University**, Teaching Assistant in Physics

RESEARCH ACTIVITIES

9/24–	**Sigma Connectivity, on assignment at Meta**, Research Scientist Redmond, Washington. Research area: *spatial audio*. Audio research for Reality Labs Research at Meta. Meta manager: Sebastian Prepeliță.
9/19–9/24	**University of California San Diego**, Ph.D. Student La Jolla, California. Center for Visual Computing; Department of Music. Research area: *computational acoustics, applied mathematics*. Mathematical modeling and numerical simulation of infinite-domain wave propagation, with applications in virtual acoustics and other wave physics domains. Committee: Albert Chern (co-chair, CSE), Miller Puckette (co-chair, Music; IRCAM), Melvin Leok (Mathematics), Stefan Bilbao (IRCAM), Sebastian Prepeliță (Meta), Shahrokh Yadegari (Music).

RESEARCH ACTIVITIES CONT.

6/23–2/24	**Meta, Reality Labs Research**, Research Scientist Intern Redmond, Washington. Research area: *spatial audio.* Audio research for Reality Labs Research at Meta. Included part-time research period (12/23–2/24). Supervisor: Sebastian Prepeliță. Team Lead: Ravish Mehra.
8/21–3/22	**Meta, Reality Labs Research**, Research Intern La Jolla, California. Included part-time research period (1/22–3/22). Research description: see above.
7/18–7/19	**Applied Research in Acoustics**, R&D Scientist Culpeper, Virginia. Research area: *underwater acoustic sensing, acoustic beamforming.* Algorithm development for underwater acoustic recognition, with applications in naval sonar systems. Team Lead: Jonathan Botts.
1/17–8/17	**University of Edinburgh**, Master's Student Edinburgh, Scotland. Acoustics & Audio Group, Reid School of Music. Research area: *computational acoustics.* Numerical algorithm development for audio-haptic devices, including the Razer Nari Ultimate headphones. Partially funded by Lofelt, a Berlin-based haptics startup acquired by Meta in 2022. Advisor: Stefan Bilbao.
1/10–9/13	**Wake Forest University**, Master's Student Winston-Salem, North Carolina. Department of Physics. Research area: *optical trapping, fluid dynamics.* Design, construction, and analysis of standing-wave optical traps for Brownian particle tracking. Published work in the Journal of the Optical Society of America B. Advisor: Keith Bonin.
8/07–8/09	**Brigham Young University**, Undergraduate Student Provo, Utah. Department of Physiology and Developmental Biology. Research area: *biophysics.* AFM imaging of artificial lipid bilayers. Advisor: David Busath.

TEACHING EXPERIENCE

As Instructor

UNCSA

SCI 1100	General Physics. Fall 2012 (1 term).

As TA

UCSD

CSE 291	Physics Simulation. Spring 2024 (1 term).
MUS 6	Electronic Music. Fall 2020 (1 term).
MUS 171	Computer Music I. Winter 2022 (1 term).
MUS 172	Computer Music II. 2021–2022 (2 terms).

WFU

PHY 113	General Physics I (Mechanics). 2009–2011 (4 terms).
PHY 114	General Physics II (E&M). Fall 2010 (1 term).

PUBLICATIONS

Manuscripts in Progress

(1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, "Wave simulations in infinite spacetime".

> *Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.*

(2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Advisors: Albert Chern and Miller Puckette.

> *This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.*

Journal Articles

(3) **C. McKell** and K. Bonin, "Optical corral using a standing-wave Bessel beam," *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

Conference Proceedings

(4) **C. McKell**, "Sonification of optically-ordered Brownian motion," In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

Master's Theses

(5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master's Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.

(6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master's Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.